

Exercise 2

chl

Exercise of Chapter 2 in the book. [Shakarchi, Stein] Complex Analysis (2003)

2-6: 1, 2, 7, 8, 9, 11, 12, 13, 14

2-7: 1

2026 年 3 月 11 日

1.

Prove that

$$\int_0^\infty \sin(x^2) dx = \int_0^\infty \cos(x^2) dx = \frac{\sqrt{2\pi}}{4}.$$

These are the Fresnel integrals. Here, $\int_0^\infty$ is interpreted as $\lim_{R \rightarrow \infty} \int_0^R$.

[Hint: Integrate the function e^{-z^2} over the path in Figure 14. Recall that $\int_{-\infty}^\infty e^{-x^2} dx = \sqrt{\pi}$.]

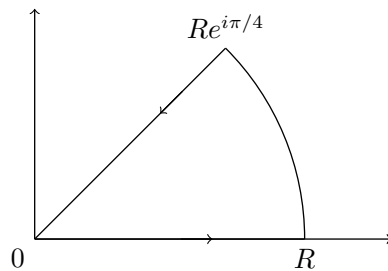


Figure 14. The contour in Exercise 1

Proof. We evaluate

$$0 = \int_{\gamma} e^{-z^2} dz = \int_0^R + \int_{arc: R \rightarrow Re^{i\pi/4}} + \int_{Re^{i\pi/4}}^0$$

We next evaluate the three integrals.

$$\begin{aligned} \left(\int_0^R e^{-x^2} dx \right)^2 &\rightarrow \left(\int_0^\infty e^{-x^2} dx \right)^2 = \int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy = \int_0^\infty \int_0^{\pi/2} e^{-r^2} r d\theta dr \\ &= \frac{\pi}{4} \int_0^\infty e^{-r^2} dr^2 = \frac{\pi}{4} \implies \int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2} \end{aligned}$$

$$\begin{aligned} \left| \int_{arc: R \rightarrow Re^{i\pi/4}} \right| &= \left| \int_0^{\pi/4} e^{-R^2 e^{2i\theta}} i R e^{i\theta} d\theta \right| \leq R \int_0^{\pi/4} |e^{-R^2 e^{2i\theta}}| \cdot |i e^{i\theta}| d\theta \\ &= R \int_0^{\pi/4} e^{-R^2 \cos(2\theta)} d\theta \leq R \int_0^{\pi/4} e^{-R^2 (1 - \frac{4\theta}{\pi})} d\theta \\ &= \frac{\pi R}{4} \int_0^1 e^{-R^2 t} dt \leq \frac{\pi R}{4} \int_0^\infty e^{-R^2 t} dt = \frac{\pi}{4R} \rightarrow 0 \end{aligned}$$

$$\begin{aligned} \int_{Re^{i\pi/4}}^0 e^{-z^2} dz &= - \int_0^R e^{-t^2 e^{i\pi/2}} e^{i\pi/4} dt = -e^{i\pi/4} \int_0^R e^{-it^2} dt \rightarrow -e^{i\pi/4} \int_0^\infty e^{-it^2} dt \\ &= -\frac{1}{\sqrt{2}}(1+i) \left(\int_0^\infty \cos(x^2) dx - i \int_0^\infty \sin(x^2) dx \right) \\ &= -\frac{1}{\sqrt{2}} \left(\int_0^\infty \cos(x^2) dx + \int_0^\infty \sin(x^2) dx \right) - \frac{i}{\sqrt{2}} \left(\int_0^\infty \cos(x^2) dx - \int_0^\infty \sin(x^2) dx \right) \end{aligned}$$

This gives

$$\begin{cases} -\frac{1}{\sqrt{2}} \left(\int_0^\infty \cos(x^2) dx + \int_0^\infty \sin(x^2) dx \right) = -\frac{\sqrt{\pi}}{2} \\ -\frac{1}{\sqrt{2}} \left(\int_0^\infty \cos(x^2) dx - \int_0^\infty \sin(x^2) dx \right) = 0 \end{cases}$$

Therefore,

$$\int_0^\infty \sin(x^2) dx = \int_0^\infty \cos(x^2) dx = \frac{\sqrt{2\pi}}{4}.$$

□

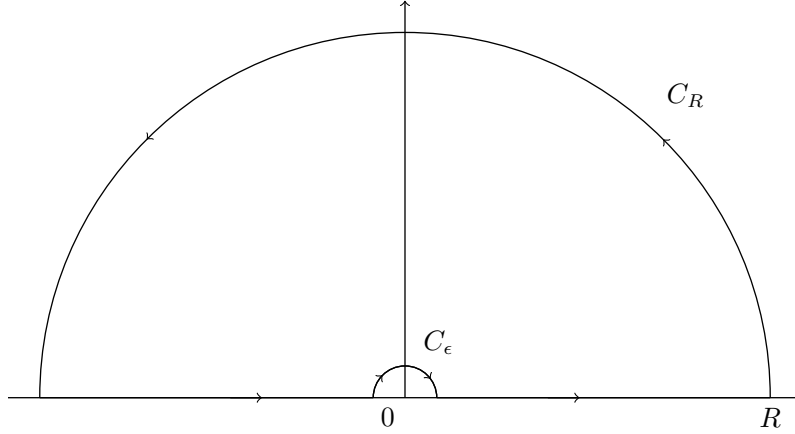
2.

Show that

$$\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}.$$

[Hint: The integral equals $\frac{1}{2i} \int_{-\infty}^\infty \frac{e^{ix}-1}{x} dx$. Use the indented semicircle.]

Proof. Consider the contour γ



Since $\frac{e^{iz}-1}{z}$ is holomorphic in $\mathbb{C} \setminus 0$, and $f(z) = \begin{cases} \frac{e^{iz}-1}{z} & z \neq 0 \\ i & z = 0 \end{cases}$, then f is holomorphic in $\mathbb{C}$,

we have

$$0 = \int_{\gamma} \frac{e^{iz}-1}{z} dz = \int_{-R}^{-\epsilon} \frac{e^{iz}-1}{z} dz + \int_{C_{\epsilon}} \frac{e^{iz}-1}{z} dz + \int_{\epsilon}^R \frac{e^{iz}-1}{z} dz + \int_{C_R} \frac{e^{iz}-1}{z} dz.$$

When $\epsilon \rightarrow 0$,

$$\left| \int_{C_{\epsilon}} \frac{e^{iz}-1}{z} dz \right| \leq \left| \int_{C_{\epsilon}} \frac{\epsilon + o(\epsilon)}{\epsilon} dz \right| = (1 + o(1)) \cdot \pi \epsilon \rightarrow 0$$

Now we evaluate the last integral.

$$\begin{aligned} \lim_{R \rightarrow \infty} \int_{C_R} \frac{e^{iz}-1}{z} dz &= \lim_{R \rightarrow \infty} \int_{C_R} \frac{\cos z}{z} dz + i \lim_{R \rightarrow \infty} \int_{C_R} \frac{\sin z}{z} dz - \lim_{R \rightarrow \infty} \int_{C_R} \frac{1}{z} dz \\ &= 0 + 0 - \lim_{R \rightarrow \infty} \int_{C_R} \frac{1}{z} dz = - \lim_{R \rightarrow \infty} \int_0^{\pi} R^{-1} e^{-i\theta} R i e^{i\theta} d\theta = -i\pi \end{aligned}$$

Thus, we have

$$\begin{aligned} i\pi &= \lim_{\epsilon \rightarrow 0} \left(\int_{-R}^{-\epsilon} \frac{e^{iz}-1}{z} dz + \int_{\epsilon}^R \frac{e^{iz}-1}{z} dz \right) = \int_{-\infty}^{\infty} \frac{e^{iz}-1}{z} dz \\ &= \int_{-\infty}^{\infty} \frac{\cos x - 1}{x} dx + 2i \int_0^{\infty} \frac{\sin x}{x} dx \end{aligned}$$

Therefore,

$$\int_{-\infty}^{\infty} \frac{\cos x - 1}{x} dx = 0 \quad \int_0^{\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}$$

□

7.

Suppose $f : \mathbb{D} \rightarrow \mathbb{C}$ is holomorphic. Show that the diameter $d = \sup_{z,w \in \mathbb{D}} |f(z) - f(w)|$ of the image of f satisfies

$$2|f'(0)| \leq d.$$

Moreover, it can be shown that equality holds precisely when f is linear, $f(z) = a_0 + a_1 z$.

Note. In connection with this result, see the relationship between the diameter of a curve and Fourier series described in Problem 1, Chapter 4, Book I.

[Hint: $2f'(0) = \frac{1}{2\pi i} \int_{|\zeta|=r} \frac{f(\zeta) - f(-\zeta)}{\zeta^2} d\zeta$ whenever $0 < r < 1$.]

Proof. $\forall 0 < r < 1$, $B(0, r) \subset \mathbb{D}$, then

$$f'(0) = \frac{1}{2\pi i} \int_{|\zeta|=r} \frac{f(\zeta)}{\zeta^2} d\zeta$$

Let $\zeta \rightarrow -\zeta$, we have

$$f'(0) = \frac{1}{2\pi i} \int_{|-\zeta|=r} \frac{f(-\zeta)}{(-\zeta)^2} d(-\zeta) = -\frac{1}{2\pi i} \int_{|\zeta|=r} \frac{f(-\zeta)}{\zeta^2} d\zeta$$

Then

$$2f'(0) = \frac{1}{2\pi i} \int_{|\zeta|=r} \frac{f(\zeta) - f(-\zeta)}{\zeta^2} d\zeta$$

Therefore,

$$2|f'(0)| \leq \frac{1}{2\pi} \left| \int_{|\zeta|=r} \frac{|f(\zeta) - f(-\zeta)|}{|\zeta|^2} d\zeta \right| \leq \frac{d}{2\pi r^2} \left| \int_{|\zeta|=r} d\zeta \right| = \frac{d}{r}$$

Let $r \rightarrow 1^-$, we have $2|f'(0)| \leq d$.

□

Proof of Equality-Linearity.

Suppose equality holds, i.e., $2|f'(0)| = d$. Then $|g'(0)| = 1$.

According to Schwarz's Lemma, if $|g'(0)| = 1$, then $g(z)$ must be a rotation: $g(z) = e^{i\theta} z$. Substituting $g(z)$ back, we have:

$$f(z) - f(-z) = de^{i\theta} z$$

This implies that the difference between $f(z)$ and $f(-z)$ is strictly linear. If we expand $f(z)$ into a Taylor series $f(z) = \sum a_n z^n$, the odd terms must satisfy:

$$2a_1 z + 2a_3 z^3 + 2a_5 z^5 + \dots = de^{i\theta} z$$

Comparing the coefficients, we find $a_1 = \frac{de^{i\theta}}{2}$ and all higher-order odd coefficients ($a_3, a_5, \dots$) are 0.

To see that all even coefficients ($a_2, a_4, \dots$) must also be 0, note that if any $a_{2k} \neq 0$, the additional variations generated by the even powers would cause the supremum of $|f(z) - f(w)|$ to strictly exceed $2|a_1| = d$, contradicting the definition of the diameter.

Therefore, $f(z)$ can only have the constant term a_0 and the linear term $a_1 z$. We conclude that $f(z)$ must be linear:

$$f(z) = a_0 + a_1 z$$

□

8.

If f is a holomorphic function on the strip $-1 < y < 1$, $x \in \mathbb{R}$ with

$$|f(z)| \leq A(1 + |z|)^\eta, \quad \eta \text{ a fixed real number}$$

for all z in that strip, show that for each integer $n \geq 0$ there exists $A_n \geq 0$ so that

$$|f^{(n)}(x)| \leq A_n(1 + |x|)^\eta,$$

for all $x \in \mathbb{R}$.

[Hint: Use the Cauchy inequalities.]

Proof. $\forall x \in \mathbb{R}$, we have

$$|f^{(n)}(x)| = \left| \frac{n!}{2\pi i} \int_{|z-x|=1/2} \frac{f(z)}{(z-x)^{n+1}} dz \right| \leq n! 2^n A(1 + |x|)^\eta \max_{|z-x|=1/2} \left(\frac{1 + |z|}{1 + |x|} \right)^\eta$$

From $|z - x| = 1/2$, we have $|x| - 1/2 \leq |z| \leq |x| + 1/2$, then if $\eta > 0$,

$$\max_{|z-x|=1/2} \left(\frac{1+|z|}{1+|x|} \right)^\eta \leq \max_{|z-x|=1/2} \left(\frac{1+|x|+1/2}{1+|x|} \right)^\eta = \left(\frac{3/2+|x|}{1+|x|} \right)^\eta \leq \left(\frac{3}{2} \right)^\eta$$

If $\eta < 0$,

$$\max_{|z-x|=1/2} \left(\frac{1+|z|}{1+|x|} \right)^\eta \leq \max_{|z-x|=1/2} \left(\frac{1+|x|-1/2}{1+|x|} \right)^\eta = \left(\frac{1/2+|x|}{1+|x|} \right)^\eta \leq \left(\frac{1}{2} \right)^\eta$$

$$\text{Let } C_\eta \triangleq \begin{cases} \left(\frac{3}{2}\right)^\eta & \text{if } \eta > 0 \\ \left(\frac{1}{2}\right)^\eta & \text{if } \eta < 0, \text{ we have} \\ 1 & \text{if } \eta = 0 \end{cases}$$

$$|f^{(n)}(x)| \leq n!2^n AC_\eta(1+|x|)^\eta \triangleq A_n(1+|x|)^\eta \quad A_n = n!2^n AC_\eta, \forall x \in \mathbb{R}$$

□

9.

Let Ω be a bounded open subset of $\mathbb{C}$, and $\varphi : \Omega \rightarrow \Omega$ a holomorphic function. Prove that if there exists a point $z_0 \in \Omega$ such that

$$\varphi(z_0) = z_0$$

and

$$\varphi'(z_0) = 1$$

then φ is linear.

[Hint: Why can one assume that $z_0 = 0$? Write $\varphi(z) = z + a_n z^n + O(z^{n+1})$ near 0, and prove that if $\varphi_k = \varphi \circ \cdots \circ \varphi$ (where φ appears k times), then $\varphi_k(z) = z + k a_n z^n + O(z^{n+1})$. Apply the Cauchy inequalities and let $k \rightarrow \infty$ to conclude the proof. Here we use the standard O notation, where $f(z) = O(g(z))$ as $z \rightarrow 0$ means that $|f(z)| \leq C|g(z)|$ for some constant C as $|z| \rightarrow 0$.]

Proof. We assume that $z_0 = 0$. If not, let $\psi(z) = z + z_0$, then $\psi^{-1} \circ \varphi \circ \psi$ is a holomorphic function on $\Omega - z_0$ with fixed point 0 and derivative 1 at 0. Therefore, we can replace φ by $\psi^{-1} \circ \varphi \circ \psi$ and assume that $z_0 = 0$.

Then we write $\varphi(z) = z + a_n z^n + O(z^{n+1})$ near 0 ($a_2 = \dots = a_{n-1} = 0, a_n \neq 0$). Let $\varphi_k = \varphi \circ \dots \circ \varphi$, if $\varphi_k(z) = z + k a_n z^n + O(z^{n+1})$, then

$$\begin{aligned}\varphi_{k+1}(z) &= \varphi(\varphi_k(z)) = \varphi(z + k a_n z^n + O(z^{n+1})) \\ &= z + k a_n z^n + O(z^{n+1}) + a_n (z + k a_n z^n + O(z^{n+1}))^n + O((z + k a_n z^n + O(z^{n+1}))^{n+1}) \\ &= z + (k+1) a_n z^n + O(z^{n+1})\end{aligned}$$

Then $\exists C > 0$ s.t. $|\varphi_k(z) - z| \leq k|a_n||z|^n + C_k|z|^{n+1}$ is bounded. Then $(\varphi_k(z) - z)^{(n)} = n!k a_n + C \rightarrow \infty$, contradiction, then $a_n = 0$ for all $n \geq 2$, and $\varphi(z) = z$ is linear. $\square$

11.

Let f be a holomorphic function on the disc D_{R_0} centered at the origin and of radius R_0 .

(a) Prove that whenever $0 < R < R_0$ and $|z| < R$, then

$$f(z) = \frac{1}{2\pi} \int_0^{2\pi} f(Re^{i\varphi}) \operatorname{Re} \left(\frac{Re^{i\varphi} + z}{Re^{i\varphi} - z} \right) d\varphi.$$

(b) Show that

$$\operatorname{Re} \left(\frac{Re^{i\gamma} + r}{Re^{i\gamma} - r} \right) = \frac{R^2 - r^2}{R^2 - 2Rr \cos \gamma + r^2}.$$

[Hint: For the first part, note that if $w = R^2/\bar{z}$, then the integral of $f(\zeta)/(\zeta - w)$ around the circle of radius R centered at the origin is zero. Use this, together with the usual Cauchy integral formula, to deduce the desired identity.]

(a)

Proof. Since $|z| < R$, we have

$$f(z) = \frac{1}{2\pi i} \int_{|\zeta|=R} \frac{f(\zeta)}{\zeta - z} d\zeta$$

For $w = R^2/\bar{z}$, $|w| = R^2/|z| > R$, then

$$0 = \frac{1}{2\pi i} \int_{|\zeta|=R} \frac{f(\zeta)}{\zeta - w} d\zeta$$

Adding up the two equations gives

$$\begin{aligned} f(z) &= \frac{1}{2\pi i} \int_{|\zeta|=R} f(\zeta) \left(\frac{1}{\zeta - z} - \frac{1}{\zeta - w} \right) d\zeta = \frac{1}{2\pi i} \int_{|\zeta|=R} f(\zeta) \frac{z - w}{(\zeta - z)(\zeta - w)} d\zeta \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(Re^{i\varphi}) \frac{z - R^2/\bar{z}}{(Re^{i\varphi} - z)(e^{i\varphi} - R/\bar{z})} e^{i\varphi} d\varphi \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(Re^{i\varphi}) \frac{|z|^2 - R^2}{(Re^{i\varphi} - z)(\bar{z}e^{i\varphi} - R)} e^{i\varphi} d\varphi \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(Re^{i\varphi}) \frac{1}{2} \frac{(Re^{i\varphi} + z)(R - \bar{z}e^{i\varphi}) - (R + \bar{z}e^{i\varphi})(Re^{i\varphi} - z)}{(Re^{i\varphi} - z)(R - \bar{z}e^{i\varphi})} d\varphi \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(Re^{i\varphi}) \frac{1}{2} \left(\frac{Re^{i\varphi} + z}{Re^{i\varphi} - z} + \frac{R + \bar{z}e^{i\varphi}}{R - \bar{z}e^{i\varphi}} \right) d\varphi \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(Re^{i\varphi}) \operatorname{Re} \left(\frac{Re^{i\varphi} + z}{Re^{i\varphi} - z} \right) d\varphi \end{aligned}$$

□

(b)

Proof.

$$\operatorname{Re} \left(\frac{Re^{i\gamma} + r}{Re^{i\gamma} - r} \right) = \frac{R^2 - r^2}{(Re^{i\gamma} - r)(R - re^{i\gamma})} e^{i\gamma} = \frac{R^2 - r^2}{R^2 + r^2 - Rr(e^{i\gamma} + e^{-i\gamma})} = \frac{R^2 - r^2}{R^2 - 2Rr \cos \gamma + r^2}$$

□

12.

Let u be a real-valued function defined on the unit disc $\mathbb{D}$. Suppose that u is twice continuously differentiable and harmonic, that is,

$$\Delta u(x, y) = 0$$

for all $(x, y) \in \mathbb{D}$.

(a) Prove that there exists a holomorphic function f on the unit disc such that

$$\operatorname{Re}(f) = u.$$

Also show that the imaginary part of f is uniquely defined up to an additive (real) constant.

[Hint: From the previous chapter we would have $f'(z) = 2\partial u/\partial z$. Therefore, let $g(z) = 2\partial u/\partial z$ and prove that g is holomorphic. Why can one find F with $F' = g$? Prove that $\operatorname{Re}(F)$ differs from u by a real constant.]

(b) Deduce from this result, and from Exercise 11, the Poisson integral representation formula from the Cauchy integral formula: If u is harmonic in the unit disc and continuous on its closure, then if $z = re^{i\theta}$ one has

$$u(z) = \frac{1}{2\pi} \int_0^{2\pi} P_r(\theta - \varphi) u(\varphi) d\varphi$$

where $P_r(\gamma)$ is the Poisson kernel for the unit disc given by

$$P_r(\gamma) = \frac{1 - r^2}{1 - 2r \cos \gamma + r^2}.$$

(a)

Proof. Let $g(z) = 2\partial u/\partial z$, we have

$$\frac{\partial g}{\partial \bar{z}} = 2\partial_{\bar{z}}(\partial_z u) = \frac{1}{2} \left(\partial_x + \frac{1}{i}\partial_y \right) \left(\partial_x - \frac{1}{i}\partial_y \right) u = \frac{1}{2} \Delta u = 0$$

This gives g is holomorphic, then according to our theorem from class, g has a primitive F with $F' = g$. If G, H satisfy $G' = H' = g$, then

$$(G - H)' \equiv 0 \implies G - H = \text{const}$$

Then $F = u + iH + \text{const}$. □

(b)

Proof. From exercise 11, we have

$$f(z) = \frac{1}{2\pi} \int_0^{2\pi} f(Re^{i\varphi}) \operatorname{Re} \left(\frac{Re^{i\varphi} + z}{Re^{i\varphi} - z} \right) d\varphi.$$

Let $z = re^{i\theta}$, $R = 1$, we have

$$\begin{aligned} f(re^{i\theta}) &= \frac{1}{2\pi} \int_0^{2\pi} f(e^{i\varphi}) \operatorname{Re} \left(\frac{e^{i\varphi} + re^{i\theta}}{e^{i\varphi} - re^{i\theta}} \right) d\varphi \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(e^{i\varphi}) \operatorname{Re} \left(\frac{e^{i(\varphi-\theta)} + r}{e^{i(\varphi-\theta)} - r} \right) d\varphi \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(e^{i\varphi}) P_r(\theta - \varphi) d\varphi \end{aligned}$$

By (a) we know that $\operatorname{Re}(f(re^{i\varphi})) = u(re^{i\varphi})$, taking the real part on both sides gives

$$u(z) = u(re^{i\theta}) = \frac{1}{2\pi} \int_0^{2\pi} u(e^{i\varphi}) P_r(\theta - \varphi) d\varphi$$

□

13.

Suppose f is an analytic function defined everywhere in $\mathbb{C}$ and such that for each $z_0 \in \mathbb{C}$ at least one coefficient in the expansion

$$f(z) = \sum_{n=0}^{\infty} c_n (z - z_0)^n$$

is equal to 0. Prove that f is a polynomial.

[Hint: Use the fact that $c_n n! = f^{(n)}(z_0)$ and use a countability argument.]

Proof. $\forall z_0 \in \mathbb{C}$, we have

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

$\exists n \in \mathbb{N}$, $a_n = \frac{f^{(n)}(z_0)}{n!} = 0 \implies f^{(n)}(z_0) = 0$. Let $A_n = \{z \in \mathbb{C} : f^{(n)}(z) = 0\}$, then $\mathbb{C} = \bigcup_{n=0}^{\infty} A_n$. Since $\mathbb{C}$ is uncountable, $\exists n_0$ s.t. A_{n_0} is uncountable.

If $f^{(n_0)}(z) \not\equiv 0$, then $\forall z^* \in A_{n_0}$, $\exists \epsilon = \epsilon_{z^*} > 0$, s.t. $\forall z \in B(z^*, \epsilon_{z^*}) \setminus \{z^*\}$, $f^{(n_0)}(z) \neq 0$. If not so, $\exists z_* \in \mathbb{C}$, s.t. $\exists \{z_n\}_{n=1}^{\infty} \rightarrow z_*$ with $f^{(n_0)}(z_n) = 0, \forall n$. Then the holomorphic $f^{(n_0)}(z) \equiv 0, \forall z \in \mathbb{C}$, contradiction.

Consider

$$\bigsqcup_{z \in A_n} B(z, \epsilon_z) \cap \mathbb{Q}[i] \quad (\text{pairwise disjoint})$$

For every $B(z, \epsilon_z) \cap \mathbb{Q}[i]$, we can take one element as a representative, then we have an injection from A_{n_0} to $\mathbb{Q}[i]$. However, $\mathbb{Q}[i]$ is countable, this gives a contradiction.

Therefore, $\forall n \geq n_0$, $f^{(n)}(z) = (f^{(n_0)})^{(n-n_0)}(z) \equiv 0$. This gives f is a polynomial of degree at most $n_0 - 1$. $\square$

14.

Suppose that f is holomorphic in an open set containing the closed unit disc, except for a pole at z_0 on the unit circle. Show that if

$$\sum_{n=0}^{\infty} a_n z^n$$

denotes the power series expansion of f in the open unit disc, then

$$\lim_{n \rightarrow \infty} \frac{a_n}{a_{n+1}} = z_0.$$

Proof. $\exists m \in \mathbb{N}$, s.t. $f(z) = \frac{g(z)}{(z-z_0)^m}$, where g is holomorphic in an open set containing the closed unit disc and $g(z_0) \neq 0$. Let $g(z) = \sum_{n=0}^{\infty} b_n z^n$, we have

$$f(z) = g(z)(-1)^m z_0^{-m} \frac{1}{(1 - \frac{z}{z_0})^m} = g(z)(-1)^m z_0^{-m} \sum_{n=0}^{\infty} \binom{m+n-1}{m-1} \left(\frac{z}{z_0}\right)^n \quad \left| \frac{z}{z_0} \right| < 1$$

Then

$$\begin{aligned} a_n &= \sum_{j=0}^n b_j (-1)^m z_0^{-m-n+j} \binom{m+n-j-1}{m-1} \\ &= (-1)^m z_0^{-m-n} \sum_{j=0}^n b_j z_0^j \binom{m+n-j-1}{m-1} \end{aligned}$$

Therefore,

$$\begin{aligned}\lim_{n \rightarrow \infty} \frac{a_n}{a_{n+1}} &= z_0 \lim_{n \rightarrow \infty} \frac{\sum_{j=0}^n b_j z_0^j \binom{m+n-j-1}{m-1}}{\sum_{j=0}^{n+1} b_j z_0^j \binom{m+n-j}{m-1}} = z_0 \lim_{n \rightarrow \infty} \frac{\sum_{j=0}^n b_j z_0^j n^{m-1} / ((m-1)!) }{\sum_{j=0}^{n+1} b_j z_0^j n^{m-1} / ((m-1)!) } \\ &= z_0 \lim_{n \rightarrow \infty} \frac{\sum_{j=0}^n b_j z_0^j}{\sum_{j=0}^{n+1} b_j z_0^j} = z_0 \cdot \frac{g(z_0)}{g(z_0)} = z_0\end{aligned}$$

□

1.

Here are some examples of analytic functions on the unit disc that cannot be extended analytically past the unit circle. The following definition is needed. Let

f be a function defined in the unit disc $\mathbb{D}$, with boundary circle C . A point w on C is said to be regular for f if there is an open neighborhood U of w and an analytic function g on U , so that $f = g$ on $\mathbb{D} \cap U$. A function f defined on $\mathbb{D}$ cannot be continued analytically past the unit circle if no point of C is regular for f .

(a) Let

$$f(z) = \sum_{n=0}^{\infty} z^{2^n} \quad \text{for } |z| < 1.$$

Notice that the radius of convergence of the above series is 1. Show that f cannot be continued analytically past the unit disc. [Hint: Suppose $\theta = 2\pi p/2^k$, where p and k are positive integers. Let $z = re^{i\theta}$; then $|f(re^{i\theta})| \rightarrow \infty$ as $r \rightarrow 1$.]

(b) * Fix $0 < \alpha < \infty$. Show that the analytic function f defined by

$$f(z) = \sum_{n=0}^{\infty} 2^{-n\alpha} z^{2^n} \quad \text{for } |z| < 1$$

extends continuously to the unit circle, but cannot be analytically continued past the unit circle. [Hint: There is a nowhere differentiable function lurking in the background. See Chapter 4 in Book I.]

(a)

Proof. Suppose $\theta = 2\pi p/2^k$, where p and k are positive integers. Let $z = re^{i\theta}$; then

$$\begin{aligned} |f(re^{i\theta})| &= \left| \sum_{n=0}^{\infty} r^{2^n} e^{i2^n\theta} \right| = \left| \sum_{n=0}^{\infty} r^{2^n} e^{i2\pi p 2^{n-k}} \right| = \left| \sum_{n=0}^{k-1} r^{2^n} e^{i2\pi p 2^{n-k}} + \sum_{n=k}^{\infty} r^{2^n} \right| \\ &= \left| C_{n,k,p,r} + \sum_{m=0}^{\infty} r^{2^{m+k}} \right| \rightarrow \infty \quad \text{as } r \rightarrow 1^- \end{aligned}$$

Therefore, the radius of convergence of f is 1, and $\forall \theta = 2\pi p/2^k$, f diverges along the radius $re^{i\theta} \rightarrow e^{i\theta}$. Since $\{e^{i\theta} : \theta = 2\pi p/2^k, p, k \in \mathbb{N}\}$ is dense in the unit circle, no point of $\mathbb{C}$ is regular for f , so f cannot be continued analytically past the unit circle. $\square$

(b)

Proof. For $\forall |z| \leq 1$, we have

$$|2^{-n\alpha} z^{2^n}| \leq 2^{-n\alpha} \quad \frac{2^{-n\alpha}}{2^{-(n+1)\alpha}} = 2^\alpha > 1$$

The power series converges uniformly on the unit disc and circle, so f extends continuously to the unit circle. However, for $z = e^{i\theta}$, we have

$$f(e^{i\theta}) = \sum_{n=0}^{\infty} 2^{-n\alpha} e^{i\theta 2^n} = \sum_{n=0}^{\infty} 2^{-n\alpha} \cos(2^n \theta) + i \sum_{n=0}^{\infty} 2^{-n\alpha} \sin(2^n \theta)$$

The real part is not differentiable at any point on the unit circle, so f cannot be analytically continued past the unit circle.

Let

$$u(\theta) = \sum_{n=0}^{\infty} 2^{-n\alpha} \cos(2^n \theta) = \cos \theta + \sum_{n=1}^{\infty} 2^{-n\alpha} \cos(2^n \theta) = \cos \theta + 2^{-\alpha} \sum_{n=0}^{\infty} 2^{-n\alpha} \cos(2^n \theta)$$

This gives $u(2\theta) = 2^\alpha(u(\theta) - \cos \theta)$, therefore if u is analytic on I , then u is analytic on $2I, 4I, \dots, \mathbb{R}$. Its Fourier coefficients are given by

$$\hat{u}(\pm 2^n) = 2^{-n\alpha}$$

Since u is analytic, $\hat{u}(n) = O(|n|^{-k})$ for all $k \in \mathbb{N}$, then $u(2^n) \sim 2^{2^n}$, this gives a contradiction. $\square$